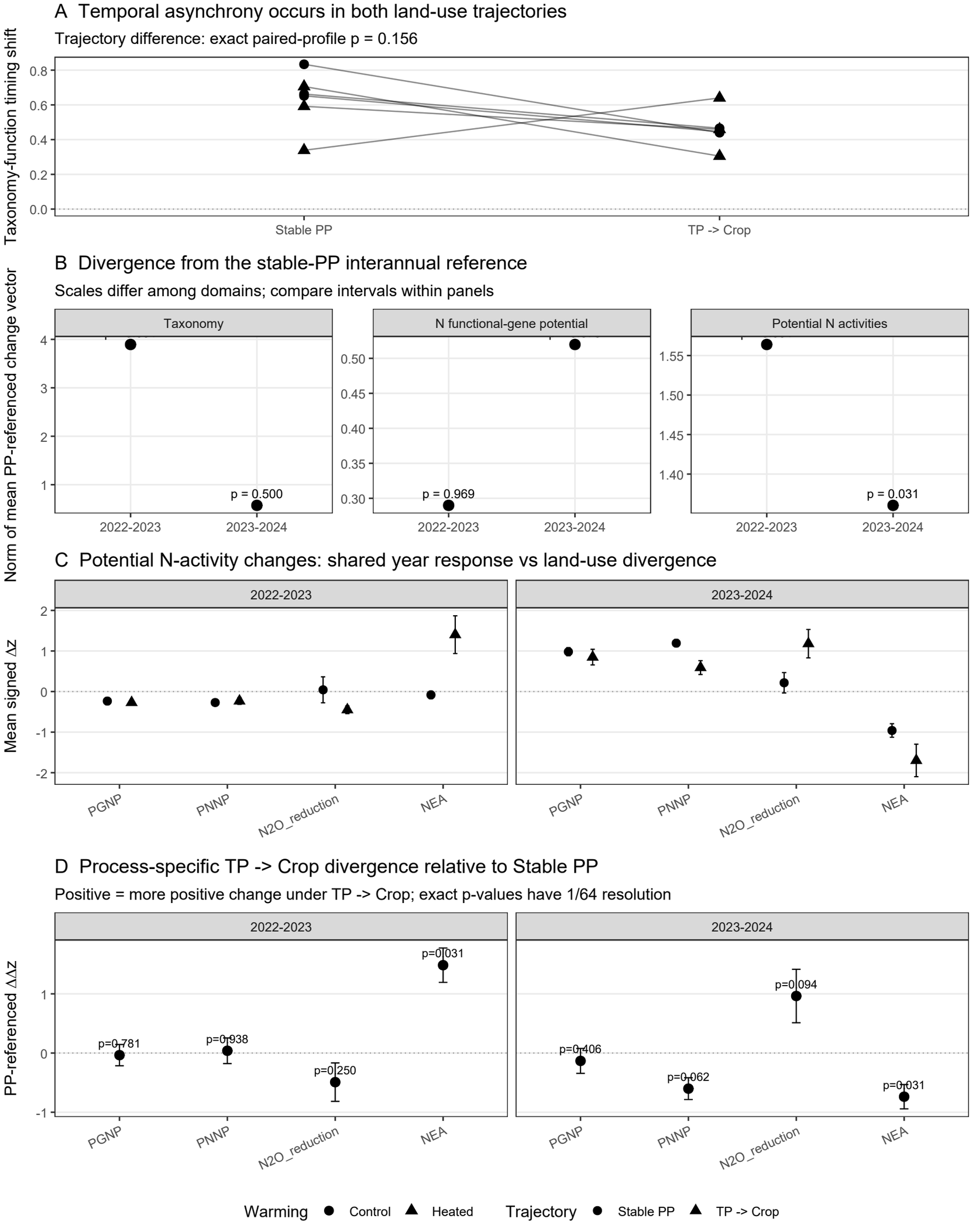
A Temporal asynchrony occurs in both land-use trajectories
Trajectory difference: exact paired-profile p = 0.156
Taxonomy-function timing shift
0.8
0.6
0.4
0.2
0.0
Stable PP
TP -> Crop
B Divergence from the stable-PP interannual reference
Scales differ among domains; compare intervals within panels
Norm of mean PP-referenced change vector
Taxonomy
N functional-gene potential
Potential N activities
p = 0.500
p = 0.969
p = 0.031
2022-2023
2023-2024
C Potential N-activity changes: shared year response vs land-use divergence
Mean signed Δz
2022-2023
2023-2024
PGNP
PNNP
N2O_reduction
NEA
D Process-specific TP -> Crop divergence relative to Stable PP
Positive = more positive change under TP -> Crop; exact p-values have 1/64 resolution
PP-referenced ΔΔz
p=0.781
p=0.938
p=0.250
p=0.031
p=0.406
p=0.062
p=0.094
p=0.031
Warming
Control
Heated
Trajectory
Stable PP
TP -> Crop